

Stacked Channel Transistors with 2D Materials: an Integration Perspective

Yun-Yan Chung¹⁺, Bo-Jhih Chou^{1,2+}, Wei-Sheng Yun¹, Chen-Feng Hsu¹, Shao-Ming Yu¹, Yu-I Chang², Chen-Yi Lee², Sui-An Chou¹, Po-Hsun Ho¹, Aslan Wei¹, D. Mahaveer Sathaiya¹, Bo-Heng Liu⁴, Chien-Wei Chen⁴, Chien-Ying Su⁴, Chi-Chung Kei⁴, Fu-Kuo Hsueh⁵, Tuo-Hung Hou^{2,5}, Wen-Hao Chang³, Jin Cai¹, Chung-Cheng Wu¹, Jeff Wu¹, Wei-Yen Woon¹, Tung-Ying Lee¹, Chao-Hsin Chien^{2*}, Chao-Ching Cheng^{1*} and Iuliana P. Radu^{1*}

¹Taiwan Semiconductor Manufacturing Company, Hsinchu, Taiwan

²Institute of Electronics, ³Department of Electrophysics, National Yang Ming Chiao Tung University, Hsinchu, Taiwan

⁴Taiwan Instrument Research Institute, ⁵Taiwan Semiconductor Research Institute, National Applied Research Laboratories, Hsinchu, Taiwan

*E-mail: chchien@nycu.edu.tw; chengcca@tsmc.com; iradu@tsmc.com *Co-first authors contributed equally.

Abstract—This report presents the first electrical demonstration of a stacked nanosheet (NS) FET with a monolayer MoS₂ channel, utilizing a typical nanosheet release process prior to high- κ metal gate deposition. We demonstrate the fabrication of **flat, two-stacked nanosheets** with a monolayer MoS₂ channel and a conformal HfO_x/TiN gate stack. The stacked nanosheets exhibit good release behavior, allowing for an integration scheme that supports gate lengths of up to 250 nm. $I_{\text{MAX}}/I_{\text{MIN}}$ ratios of $\sim 1\text{E}5$ and a subthreshold swing of ~ 220 mV/dec are reported. Additionally, for the first time, the integration of stacked two NS WSe₂ and two NS MoS₂ in the same structural FET is demonstrated using the NS release and high- κ metal gate conformal deposition process. The two channel materials represent typical PMOS and NMOS 2D materials, respectively.

INTRODUCTION

Nanosheet (NS) transistor architectures are crucial for next-generation logic technology due to their enhanced performance capabilities [1]. In recent years, the semiconductor community has been exploring monolayer or few-layer transition metal dichalcogenides (TMDs), also known as “2D materials,” as potential channels for very scaled device sizes. The scaling potential stems from the capability to form monolayer thin channels, hence scaling gate length to sub-10nm dimensions [2]. In addition, contact resistance has been reported to have only weak dependence on contact length down to $\sim 15\text{nm}$ [3].

While 2D materials possess interesting electrostatic properties, achieving high mobility or current density has proven to be challenging. It is increasingly evident that in order to attain performance comparable to silicon, a device with a similar or larger effective width is necessary. This necessitates the integration of 2D materials channels into stacked NS structures. By implementing a c-type contact [4] in conjunction with channel stacking, models indicate that the device/cell current would increase more rapidly than capacitance with the number of nanosheets in the transistor, as illustrated in **Fig. 1**. The underlying model has been detailed elsewhere [5]. The acceleration of cell current relative to cell capacitance with an increasing number of nanosheets leads to enhanced circuit speed or reduced power consumption.

For a single nanosheet with a 2D material channel, a promising current density (I_{MAX}) of $\sim 400 \mu\text{A}/\mu\text{m}$ with a subthreshold swing of ~ 86 mV/dec has been reported with monolayer (1L) MoS₂ [6]. However, for stacked NS transistors, only successful demonstrations of critical modules and a gate-all-around (GAA) NS architecture have been shown [4]. The difficulty in fabricating

NS-like transistors arises from the thinness of the channel ($\sim 0.7\text{nm}$) and the lack of insight into wet and dry etch chemistry and interactions for the 2D materials. In this paper, we introduce an integration scheme that mitigates both challenges by using carefully chosen supporting/protection layers. However, it is worth emphasizing that deeper fundamental work in etch selectivity and mechanical modeling of the processes is required. It is also worth noting that the current integration demonstration lacks a key known element for increased device performance, the use of Sb contacts. Conformal ALD processes for Sb deposition are desired for this.

RESULTS AND DISCUSSIONS

A. Single NS: monolayer MoS₂ & monolayer WSe₂

Unlike TMD films transferred onto a gate electrode in planar devices, suspended NS allows for depositing a symmetric gate stack in a gate-all-around configuration. The interlayer of the gate stack is expected to have an impact not only on subthreshold swing (SS) but also on mobility through remote phonon scattering. Models project that interlayers such as SiON_x should allow for higher mobility. In a simplified short loop flow without channel release (refer to **Fig. 2**), we test this model. Two interlayers deposited directly on the 1L-MoS₂ sheet in devices are compared: low temperature HfO₂ and higher temperature SiON_x. Transfer characteristics of devices with these two interlayers and an HfO_x/TiN gate stack are plotted in **Figs. 3a and b**. The devices with HfO₂ interlayer exhibit a higher median I_{MAX} of $\sim 100 \mu\text{A}/\mu\text{m}$, a larger on-off ratio of ~ 9 orders, as well as better subthreshold swing (SS) of ~ 190 mV/dec at $V_{\text{DS}} = 1$ V than those with the SiON_x interlayer. The difference is attributed to chemical interactions/damage to the channel from the second process, as expected, even though the remote phonon scattering is lower. Despite the higher SS, the SiON_x interlayer enables hysteresis-free operation and lower variability. This is linked to the different types of defects formed and present at the two types of interfaces and their respective energy alignment with the defects in the bulk of the gate dielectric.

The study focused on single-layer WSe₂, but the yield of suspended NS is not as good as with 1L-MoS₂ sheets (refer to **AFM images in Fig. 4**). It is widely recognized in the field that WSe₂ has weaker mechanical strength compared to MoS₂. Mechanical forces generated during wet cleaning and patterning steps can significantly impact monolayer TMD films, resulting in a lower yield of suspended nanosheets. To address this issue, we introduce thin dielectric films below and above the 1L-WSe₂ channel, as depicted in **Fig. 5**. The method offers advantages such as increased selectivity in various etching and wet cleaning processes, as well as a wider process window for high- κ dielectric deposition. SEM images in **Fig. 6** demonstrate the improved yield

in suspending WSe₂ sheets. For a TEM cross-section and EDX elemental mapping after high- κ metal gate deposition to confirm the planarity of the channel, please refer to Fig. 7.

B. Stacked NS process module optimization.

A critical module for the performance of stacked nanosheet devices is the formation of high-coverage 'C-type' metal contacts to the monolayer TMD channel. TEM images in Figs. 8a and b show that the profile of the 'C-type' metal contact depends on the material selection in the stacked structure, especially on the protection layer, as well as on the surface preparation in the source/drain (S/D) before metal contact. For the electrical device demonstration here, a simplified flow with only edge contact is implemented. The contact metal still requires conformal deposition, but only to the entire vertical stack. ALD-TiN is used as the contact metal here, although we acknowledge that other metals such as Sb would be preferred. The TiN metal contributes to the anchoring of the nanosheets. The removal of the TiN from the channel region is executed using atomic layer etching (ALE), followed by the NS release process. The suspended MoS₂ stacked sheets are built with enough sheet-to-sheet spacing to enable non-scaled gate metal filling, as shown in the SEM image in Fig. 9a.

High- κ /metal-gate deposition processes can introduce uniaxial stress, leading to nanosheet bending (smiling). An example TEM image is displayed in Fig. 9b. While the uniaxial stress might have little impact on the electrical properties, nanosheet deformation results in weaker anchoring at the contacts and can impact yield. The NS bending can be mitigated through ALD process optimization. The TEM cross-section of 2-stacked NS with 1L-MoS₂ channel after metal gate optimization is displayed in Figs. 10a-c. Note that the MoS₂ sheets are flat without visible stress-induced effects, without compromising the conformality of the HfO_x and TiN gate stack. These results are further confirmed by elemental mapping plotted in Figs. 10d-i.

C. Stacked NS transistor performance.

Fig. 11a and b display SEM images of stacked NS devices with single (1 $\times$) finger and multiple (5 $\times$) fingers. Both cell layouts allow for NS release and subsequent conformal high- κ metal gate at the gate length (L_G) = 100 nm. The corresponding transfer characteristics are plotted in Fig. 12. The median performance shows a relatively low $I_{MAX} \sim 4 \mu A$ /each (lower than current density in planar back-gates devices using Sb contact), on-off ratio ~ 5 orders at $V_{DS} = 1$ V, with a threshold voltage (V_{TH-min}) ~ 0 V. The low current density is attributed to the formation of only edge contact to the nanosheets and the non-ideal metal (TiN) used in this process. The best-performing single- and multi-finger devices are compared in Fig. 13. A higher I_{MAX} with a lower hysteresis is observed for the multi-finger devices, while their output characteristics display better current linearity at low V_{DS} . These findings may indicate worse edge contact in single-finger devices, possibly due to insufficient selectivity during active area patterning.

SEM images in Fig. 14 display the L_G -dependent stacked nanosheets. Nanosheets can span up to 250 nm without collapsing. This stems from the enhanced mechanical stiffness in the NS channel sandwiched with the protection layer. As L_G scales, the I_{ON} at the same gate overdrive increases slightly but only at high V_{DS} . See Fig. 15 for these plots. The lack of current scaling indicates that the I_{ON} performance is limited by the contact resistance, not surprising with edge-only contact. This is the first study to report functional stacked NS devices with a monolayer

MoS₂ channel fabricated with channel release and GAA processes. However, performance improvement and a longer process flow using C-type contacts are required.

D. CFET-like structure of Stacked TMD nanosheets.

Complementary FET (CFET) featuring NMOS and PMOS stacked on top of each other is seen as a viable option for future technology nodes [7]. When it comes to 2D materials, MoS₂ has shown the best NMOS performance, while WSe₂ is considered a promising candidate for PMOS. In this study, we combine these two materials in a CFET-like structure by employing two stacked nanosheets for each NMOS and PMOS transistor. As shown in the TEM image in Fig. 16, four TMD channels are stacked, with the top two comprising monolayer WSe₂ and the bottom two consisting of monolayer MoS₂. Each TMD channel is surrounded by protective dielectrics and separated from one another by a sacrificial (SAC) dielectric layer. Elemental mapping using EDX helps in identifying the different layers (refer to Fig. 16). Raman spectra in Fig. 17 confirm that the stacking and deposition of multiple dielectrics do not alter the monolayer TMDs' structure.

It is worth noting that in this stacked CFET structure, the protection/supporting layer is crucial as it enables the TMD channel to withstand large mechanical forces and chemical attacks during the NS release. As shown in the SEM image in Fig. 18, successful release of both WSe₂ and MoS₂ stacked nanosheets is achieved after selective removal of the SAC layer. With subsequent high- κ and metal-gate formation, each NS becomes distinct in the SEM image (see Fig. 19a). The corresponding TEM image in Fig. 19b verifies the successful formation of the top two WSe₂ sheets and the bottom two MoS₂ sheets that are wrapped with a conformal HfO_x/TiN gate stack in the structure. To our knowledge, this is the first demonstration of a CFET-like NS architecture with four stacked monolayer TMD channels.

CONCLUSION

2D materials have good mechanical properties, and even in their monolayer form, they can withstand mechanical forces present in the nanosheet release process. The yield of the nanosheet structures is improved by introducing dielectric layers, which serve the combined role of mechanical support and protection against chemical attacks. Through careful optimization of the strain from the high- κ and metal gate stack, two flat nanosheets, each with a monolayer MoS₂ channel surrounded by a conformal HfO_x/TiN, have been demonstrated. Transistors fabricated in this manner displayed I_{MAX} of approximately 4 μA /each, limited by the contact scheme, while the on-off current ratio reached around 1E5. Additionally, monolayer MoS₂ and WSe₂ materials were stacked as multiple n- and p-channels to demonstrate a CFET-like NS structure via channel release and conformal single gate stack deposition. This is the first study to demonstrate electrically active transistor action with two-stacked MoS₂ NS devices. The work highlights the necessity of developing process steps customized for these channel materials.

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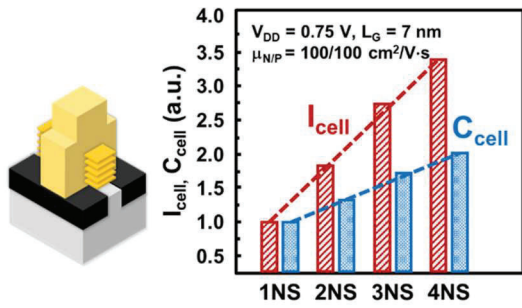


Fig. 1. Sketch of a stacked TMD NS channel transistor. Normalized I_{cell} and C_{cell} performance from TCAD modeling for the stacked NS inverter cell. I_{cell} is expected to increase faster than C_{cell} as the number of nanosheets increases and R_c decreases accordingly.

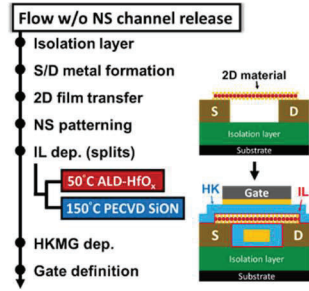


Fig. 2. Key steps of the short-loop process flow without channel release. The main parameter tested is the deposition conditions of the interfacial layer.

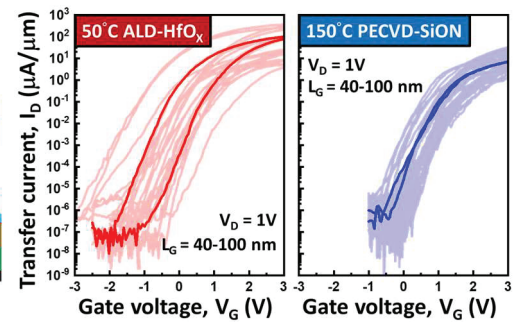


Fig. 3. Bi-directional I_D - V_{GS} sweeps were conducted for 1L-MoS₂ single NS n-FETs with two interfacial layer (IL) conditions in the gate stack. The saturated color lines represent the median performance.

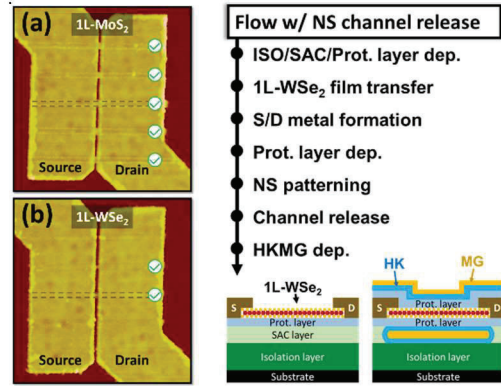


Fig. 4. AFM images of (a) 1L-MoS₂ and (b) 1L-WSe₂ transferred onto S/D metal.

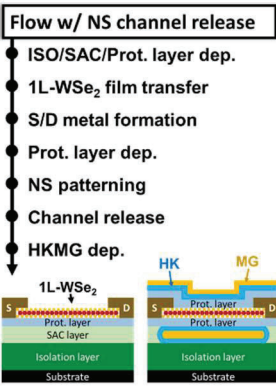


Fig. 5. Key process flow involve channel release. The inclusion of a protection layer around the 1L-WSe₂ channel enhances mechanical stability and provides chemical protection.

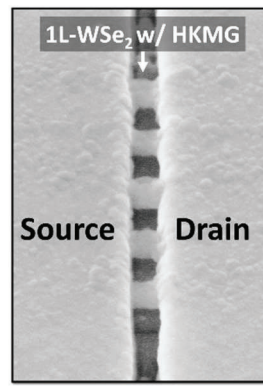


Fig. 6. Tilted SEM image of multiple devices with 1L-WSe₂ NS after the HKMG process. All devices are successfully suspended.

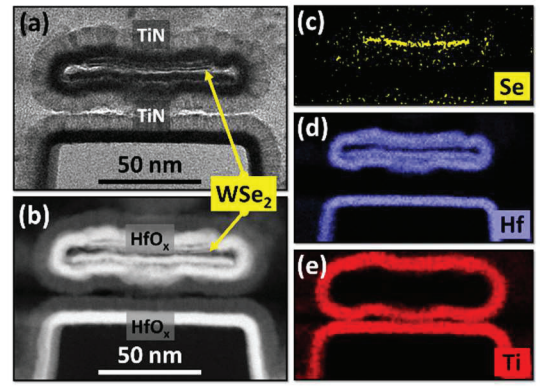


Fig. 7. (a) and (b) show TEM images of 1L-WSe₂ NS after HKMG deposition. (c)-(e) display EDX mapping of Se, Hf, and Ti confirming the presence of the 1L-WSe₂ NS channel with GAA HfO_x/TiN.

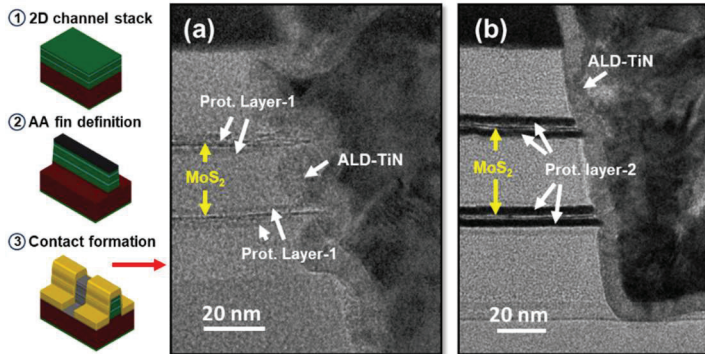


Fig. 8. Contact formation. TEM images of source-drain ALD-TiN contact formation using two types of protection materials. (a) C-type contact is shown with protection layer-1, while edge contact is obtained with protection layer-2.

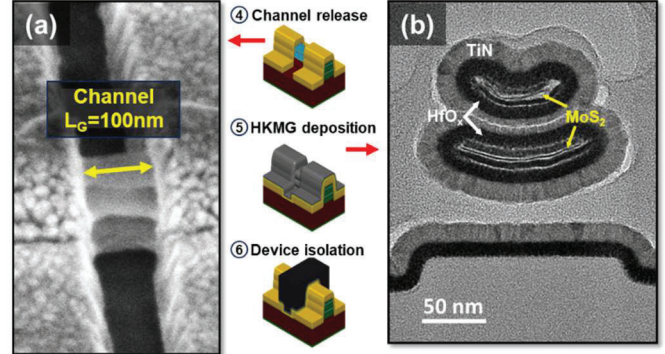


Fig. 9. Channel release and HKMG deposition. (a) Tilted SEM image of stacked NS after channel release. (b) TEM image displays one example of stacked NS with a bent channel profile after HKMG formation.

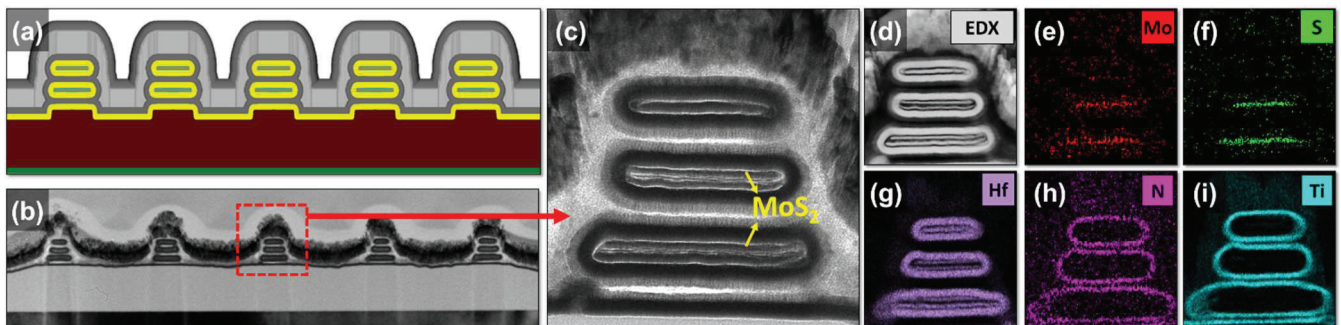


Fig. 10. HKMG deposition. (a) Process software model and (b) TEM cross-section of 2-stacked NS FET with 5 fingers. (c) Zoom-in TEM image and (d)-(i) EDX mappings of stacked 1L-MoS₂ NS channel with HKMG. 1L-MoS₂ in two stacked nanosheets after a conformal HKMG film scheme showing no bending after improved process.

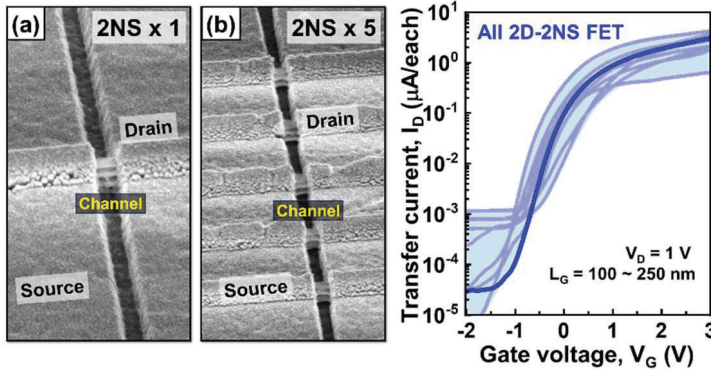


Fig. 11. Tilted SEM images of 2-stacked NS after channel release for (a) 1-finger and (b) 5-finger channels. No sagging of the NS is observed at $L_G = 100$ nm.

Fig. 12. Transfer curves of stacked 1L-MoS₂ NS with L_G ranging from 100 to 250 nm. The saturated color curve represents the median performance.

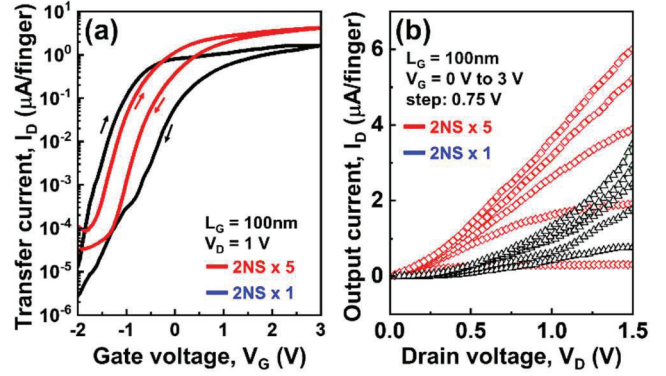


Fig. 13. (a) Transfer and (b) Output characteristics for 2-stacked NS with 1 and 5 fingers respectively. Multi-finger NS device exhibits higher I_{ON} and lower hysteresis. Current density is limited by the edge contact scheme used here.

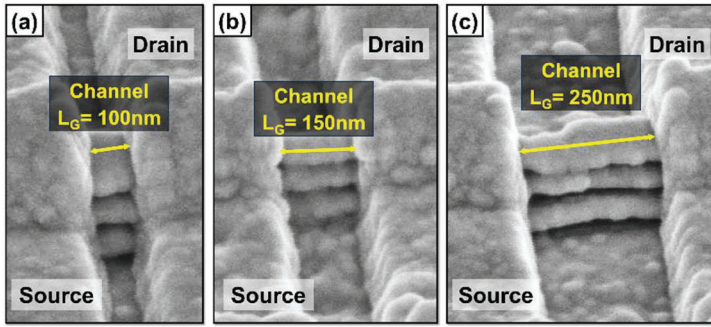


Fig. 14. Tilted SEM images of 2-stacked NS after HKMG at gate lengths of: (a) 100 nm, (b) 150 nm, and (c) 250 nm. All nanosheets display suspended behavior with relatively little sagging.

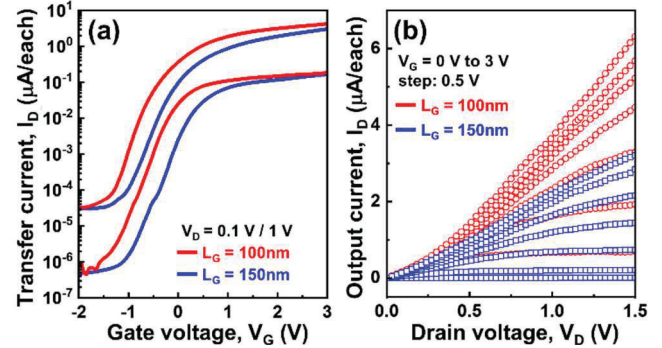


Fig. 15. (a) I_D - V_{GS} and (b) I_D - V_{DS} characteristics for 2-stacked NS FET with $L_G = 100$ nm and 150 nm. The data for the 5-finger NS channel device are compared.

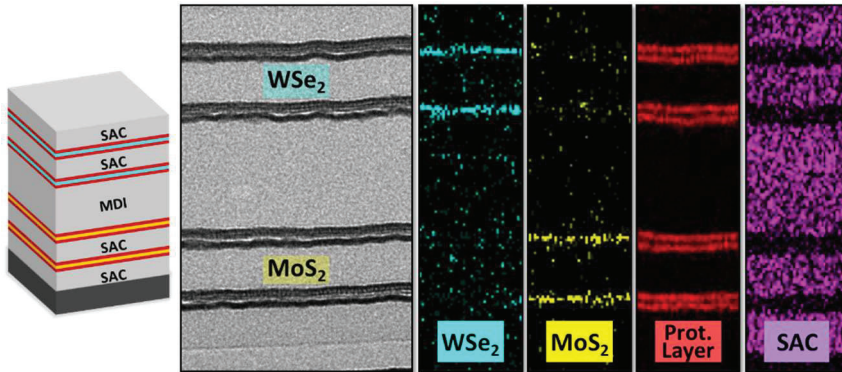


Fig. 16. Preparation of stacked TMD film scheme for complimentary FET demonstration. The TEM cross-section shows 2-stacked MoS₂ and 2-stacked WSe₂ films. EDX mapping confirms TMD layers are well-covered by the protection layer and further isolated by the SAC layer.

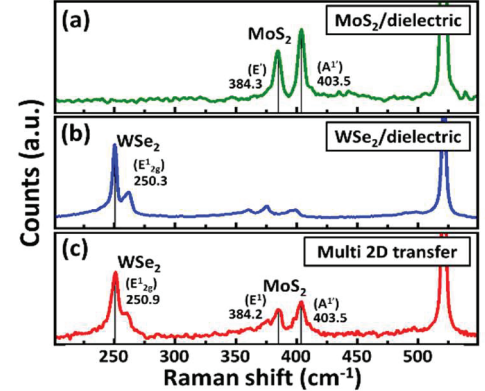


Fig. 17. The Raman spectra of (a) 1L-MoS₂ and (b) 1L-WSe₂, and (c) stacked MoS₂ and WSe₂ films in the structure in Fig. 16.

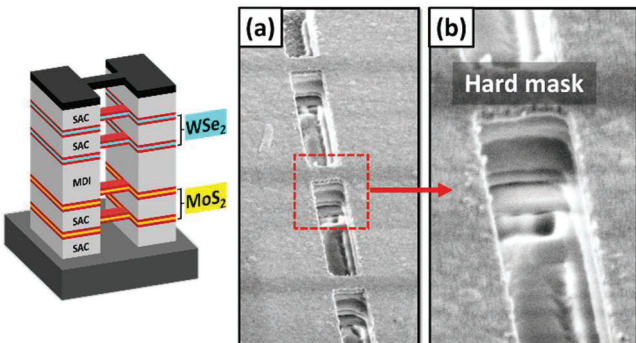


Fig. 18. (a) Tilted SEM and (b) zoom-in images of two WSe₂ channels and two MoS₂ channels after channel release.

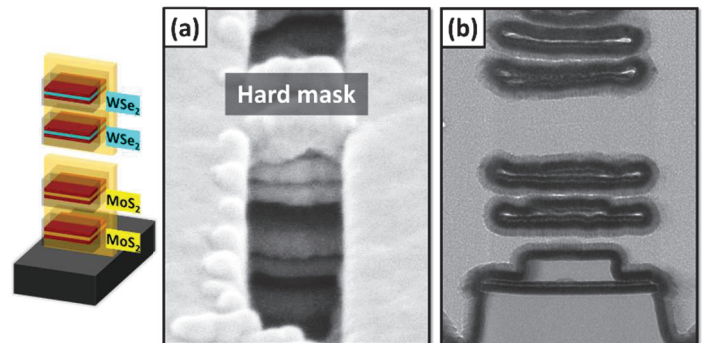


Fig. 19. (a) Tilted SEM and (b) TEM images of stacked WSe₂ and MoS₂ NS channels after HKMG deposition. Four stacked flat nanosheets and good conformity of the HfO_x/TiN gate scheme are demonstrated.